

Ying Siu (Sheryl) Liang, Ph.D.

Robotics Software Engineer

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Nationality/Location: Vienna, Austria

Summary

I specialize in human-centric AI and intuitive robotic systems and develop generalisable solutions for real-world scenarios, making complex robots accessible to non-technical users.

Keywords: end-user robotics, cognitive robotics, social robotics, human-robot interaction

Education

- 2016 - 2019 **Ph.D. in Robotics**, *Université Grenoble Alpes*, France.
Thesis: *End-User Robot Programming in Cobotic Environments*.
- 2015 - 2016 **Master of Science (M.S.) in Informatics**, *Grenoble INP - ENSIMAG*, France.
Specialisation: Artificial Intelligence and Web.
- 2008 - 2012 **Integrated Master of Science (MSci) in Mathematics and Computer Science**,
Imperial College London, United Kingdom, *First Class Honours*.

Experience

- 2022 - 2026 **Robotics Software Engineer**, *Aeolus Robotics*, Vienna, Austria.
Robot Behaviour Planning and Execution (C++, Python)
- Developed solutions for robot reasoning in real-world, user-centric scenarios such as reactive strategies to handle environment changes including failure-handling and recovery behaviours for unpredictable edge-cases.
 - Designed performance metrics and rigorous lab testing protocols to ensure robots consistently met and exceeded product requirements.
 - Drove cross-functional integration with Manipulation, Vision, and Navigation teams and ensured resilient robot behaviours with live troubleshooting.
- 2020 - 2021 **Research Scientist**, *A*STAR Institute for High Performance Computing*, Singapore.
CollabAI project in Social and Cognitive Computing (Python, Clojure)
- Developed a novel object permanence model for robot belief management that tracks invisible displacements in dynamic environments and outperformed state-of-the-art models in spatial reasoning and situational awareness. [1,2]
 - Led the Cognitive Systems Group in developing and integrating the ICARUS architecture. Integrated high-level reasoning components to combine NLU, perception, and manipulation modules and to drive goal-oriented behaviours on Universal Robots (UR5, UR16e). [3]
- Jun-Sep 2019 **Research Intern**, *A*STAR Institute for Infocomm Research*, Singapore.
Inferring the Geometric Nullspace for Robot Skills from Human Demonstrations (C++) [4]
- 2016 - 2019 **Ph.D. in Robotics**, *Grenoble Computer Science Lab*, Univ. Grenoble Alpes, France.
End-User Robot Programming in Cobotic Environments (C++, JavaScript/Polymer) [5-9]
- Pioneered iRoPro, an interactive robot programming framework that lets end-users teach robots new actions by demonstration, enabling them to tackle previously unseen, complex tasks on the fly. Engineered a full-stack robotics system by fusing advanced object perception (point cloud segmentation), manipulation, motion planning (MoveIt), behavior planning (PDDL), and human-computer interaction into a seamless, end-to-end solution.
 - Designed and executed human-robot interaction studies to rigorously evaluate user acceptance, usability, and overall system performance.

- Jul-Dec 2017 **Research Intern**, *Human-Centered Robotics Lab*, Univ. of Washington, USA.
End-User Programming of Goals and Actions for Robotic Shelf Organisation (C++) [8]
- Feb-Jun 2016 **Research Intern**, *Grenoble Computer Science Lab*, Univ. Grenoble Alpes, France.
Robot Programming by Demonstration for Non-Experts - Proof of Concept (Python) [10]
- 2012 - 2015 **Software Engineer**, *Deutsche Bank Global Technology*, London, United Kingdom.
- Implemented features for data access from external sources, allowing users to query and update information directly within an OLAP Cube through a web application (C#, HTML/JavaScript).

Skills

- Programming C++, Python, Clojure, C#, SQL, PHP, HTML/JavaScript
- Robotic ROS, RVIZ, OpenCV, PDDL
- Frameworks Aeolus Robot, Universal Robot, Baxter Robot, Fetch Robot, QTRobot
 Ubuntu Linux (CLI), Jenkins (CI/CD), Docker, Git, Visual Studio Code
- Spoken German (native), English (bilingual), French (C2), Chinese (B2), Spanish (A1)

Attended Courses

- 2021 IS5452 Affective Computing, National University of Singapore, Singapore
- 2018 ROS Summer School, FH Aachen, Germany

Additional Responsibilities

- 2023-2026 **Math and French tutor**, *Private tutor for high school students*, Austria.
- 2021 **Team lead**, *Project on Engagement Detection in Virtual Classes*, Singapore.
- 2021 **Co-coordinator**, *NIE-A*STAR Workshop for AI in Education*, Singapore.
- 2018 **Mentor of undergraduate interns**, *UI design for iRoPro*, Grenoble, France.
- 2017 **Teaching assistant**, *Introduction to Databases*, Grenoble, France.
- 2017 - 2023 **Reviewer**, *HRI'23, IROS'21, ICRA'18, '19, '21, ACII'21, RSS'19, RO-MAN'19, HRI'18.*

Volunteering

- 2021 **Elementary math tutor**, *South Central Community Service Centre*, Singapore.
- 2020 - 2021 **Trishaw pilot for nursing homes**, *Cycling Without Age*, Singapore.
- 2013 - 2015 **GCSE math tutor**, *Volunteering Matters*, London, United Kingdom.

Awards and Honours

- 2016 - 2019 French Ministry Ph.D. research grant by MSTII Doctoral School (ED MSTII)
- Mar 2018 Prize for best presentation at the LIG Ph.D. day
- Jul 2017 IDEX International Mobility Grant to collaborate with the Univ. of Washington
- Jun 2017 Prize for best presentation at the ED MSTII Ph.D. day

Hobbies

Basketball, Cycling, Table tennis, Snowboarding, Boardgames

Publications and Other Communication

- [1] Y.S. Liang, C. Zhang, D. Choi, K. Kwok (2021). **Improving Object Permanence using Action Annotations and Reasoning**. Intl. Conf. on Intelligent Robots and Systems (IROS). Virtual (Oral): IEEE/RSJ Press.
 - [2] Y.S. Liang, D. Choi, K. Kwok (2021). **Maintaining a Reliable World Model using Action-aware Perceptual Anchoring**. Intl. Conf. on Robotics and Automation (ICRA). Xi'An, China (Oral): IEEE Press.
 - [3] D. Choi, W. Shi, Y.S. Liang, K.H. Yeo, J.J. Kim (2021). **Controlling Industrial Robots with High-level Verbal Commands**. Intl. Conf. on Social Robotics (ICSR). Singapore: IEEE/RSJ Press.
 - [4] C. Cai, Y.S. Liang, N. Somani, Y. Wu (2020). **Inferring the Geometric Nullspace of Robot Skills from Human Demonstrations**. Intl. Conf. on Robotics and Automation (ICRA). Virtual: IEEE Press.
 - [5] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2021). **iRoPro, an interactive Robot Programming Framework**. Intl. Journal of Social Robotics (IJSR). Springer-Verlag.
 - [6] Y.S. Liang (2019). **End-User Robot Programming in Cobotic Enviroments**. PhD thesis. Grenoble, France.
 - [7] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2019). **End-User Programming of Low- and High-Level Actions for Robotic Task Planning**. Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). New Delhi, India (Oral): IEEE Press.
 - [8] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty, M. Cakmak (2018). **Simultaneous End-User Programming of Goals and Actions for Robotic Shelf Organization**. Intl. Conf. on Intelligent Robots and Systems (IROS). Madrid, Spain (Oral): IEEE/RSJ Press.
 - [9] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **Evaluation of a Robot Programming Framework for Non-Experts using Symbolic Planning Representations**. Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). Lisbon, Portugal (Poster): IEEE Press.
 - [10] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **A Framework for Robot Programming in Cobotic Environments: First User Experiments**. Intl. Conf. on Mechatronics and Robotics Engineering (ICMRE). Paris, France (Oral): ACM.
- May 2018 **Spotlight talk**, *A Robot Programming Framework for Non-Experts*, Robots in Human Environments (RHUM) Workshop, Grenoble, France.
- Jun 2017 **Oral presentation**, *Robot Programming by Demonstration*, Affects, Artificial Companions and Interactions (ACAI) Working Group, Paris, France.